

Chapter 5: Trigonometric Functions — Unit Circle Approach

Name: _____

Date: _____

Instructions: Answer all questions. Show complete work and justify your steps. Give *exact* values (in terms of π and radicals) unless otherwise specified. Clearly indicate whether angles are in degrees or radians. For graphing, label amplitude, period, midline, phase shift, and asymptotes as appropriate.

Each problem begins on a new page.

1. For each expression, state whether it is positive, negative, zero, or undefined. Justify briefly using quadrant information or a unit-circle argument. (10)
- $\sin 247^\circ$, $\cos(-112^\circ)$, $\cot 128^\circ$, $\csc 88^\circ$.

2. For how many values of θ with $0 \leq \theta \leq 720^\circ$ is $\cos \theta = \frac{\sqrt{3}}{2}$? List the solutions in degrees. (10)

3. Let $f(x) = 1 - 2\sin(3x - \frac{\pi}{6})$. Determine the amplitude, period, phase shift, midline, and range of f . Sketch one full period and label intercepts and extrema. (12)

4. How many points with x -coordinate between 0 and 100 (inclusive) do the graphs of $y = \sin x$ and $y = \cos x$ intersect? Explain your reasoning. (10)

5. Consider $g(x) = 2 \tan(3x - \pi)$. State the domain in set-builder or descriptive form (as all real numbers except a set), the range, the period, and list two consecutive vertical asymptotes. (12)

6. Let $y = \arctan x$. Express $\sin y$ and $\cos y$ entirely in terms of x (no trigonometric functions remaining). State any domain restrictions used in your derivation. (14)

7. Evaluate each exactly, giving answers in radians. (10)

(a) $\arccos\left(-\frac{\sqrt{3}}{2}\right)$

(b) $\arcsin\left(-\frac{1}{2}\right)$

(c) $\arctan(\sqrt{3})$

(d) $\arccos(0)$

8. While finding the sine of a certain angle, an absent-minded professor failed to notice that the calculator was not set to the correct angular mode. He was lucky to get the right answer. Find the two smallest positive values of x such that the sine of x degrees equals the sine of x radians; that is, (12)

$$\sin(x^\circ) = \sin x.$$

Give numerical answers accurate to three decimal places.

9. Solve for real θ :

$$2 \tan\left(5\theta + \frac{\pi}{6}\right) = -\sqrt{3}.$$

(10)

Give the general solution and list the solutions in $[0, 2\pi)$.

10. A block attached to a vertical spring oscillates so that its distance d (in meters) from the ceiling varies sinusoidally between 0.9 m and 1.7 m. It takes 1.2 seconds to travel from the high point to the low point. At $t = 0$ the block is at a high point. (12)
- (a) Write a formula $d(t)$ modeling the motion.
- (b) How far from the ceiling is the block at $t = 2.7$ seconds?